A Vector Refresher

Trim Ch 11, S. 11.1,11.3,11.4,11.6

MTE 203 – Advanced Calculus

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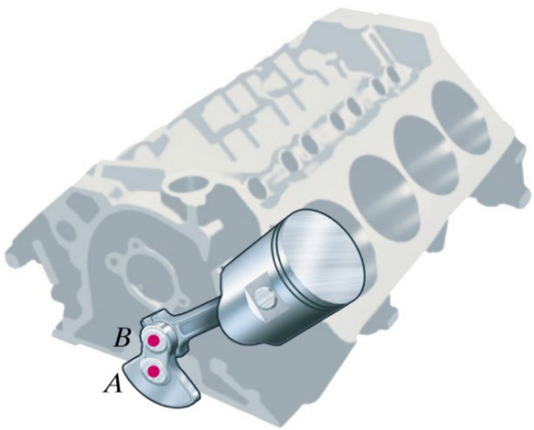
Vectors

Scalars represent physical quantities that have associated with them only a magnitude (e.g. temperature, density, area, moment of inertia, speed) and can be represented by real numbers.

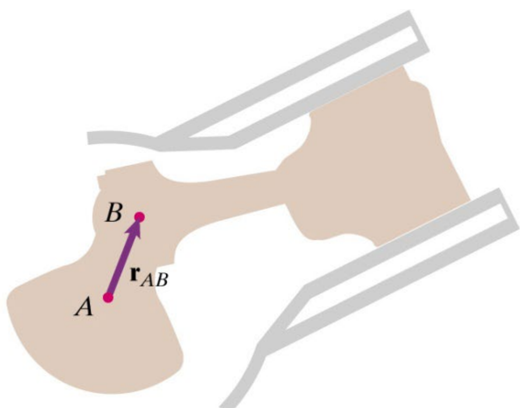
Vectors represent physical quantities that have associated with them both magnitude and direction (e.g. velocity, acceleration, force).

Reference: Trim, Ch.11, Section 11.3

Vector Definitions



Two Points A and B of a mechanism



Vector $\vec{r}_{AB}$ from A to B

Vector:

Directed line segment with both direction and length ($\vec{r}_{AB}$).

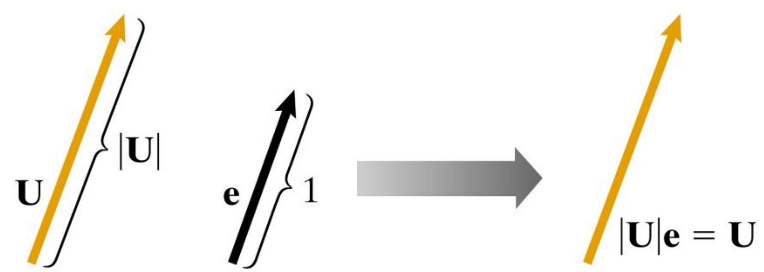
Notation:

We will use an arrow above the vector name (e.g. $\vec{U}$, $\vec{V}$). Note that your textbook uses the bold notation.

Equality of vectors:

Two vectors are equal if and only if they have the same length and direction. Their points of application are irrelevant.

Unit Vector:



A vector is said to be a unit vector if it has a length equal to one unit.

$$\hat{e} = \frac{\vec{U}}{|\vec{U}|} = \frac{(U_x, U_y, U_z)}{\sqrt{{U_x}^2 + {U_y}^2 + {U_z}^2}}$$

Components of a Vector in Arbitrary Directions

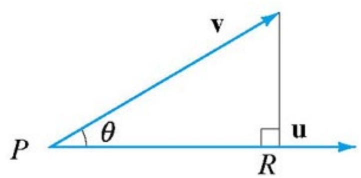
The components of a vector $\vec{v}$ in an arbitrary direction $\vec{u}$ is,

$$\left|\overrightarrow{PR}\right| = \vec{v} \cdot \hat{u}$$

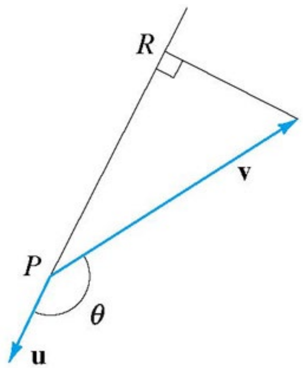
which can also be written as,

$$\left|\overrightarrow{PR}\right| = |\vec{v}||\hat{u}| \cos\theta$$

where $\hat{u}$ is the unit vector in the direction of $\vec{u}$.



$\left|\overrightarrow{PR}\right|$ is the component of the vector $\vec{v}$ in the direction of $\vec{u}$ ($\theta < 90°$)



Situation where the component of the vector $\vec{v}$ in the direction of $\vec{u}$ is negative ($\theta > 90°$)

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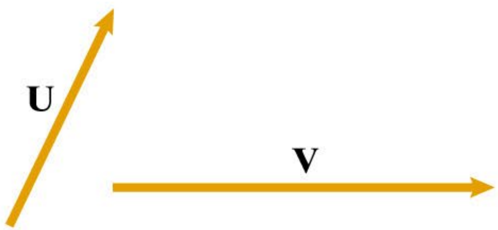
Reference: Trim, Ch.11, Section 11.3

Addition of Vectors

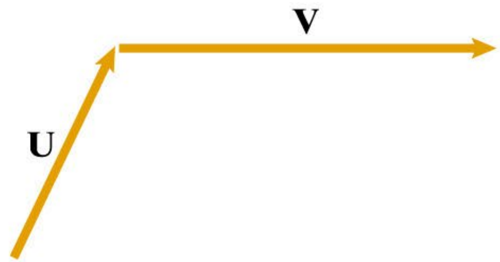


1. Two Vectors $\vec{U}$ and $\vec{V}$

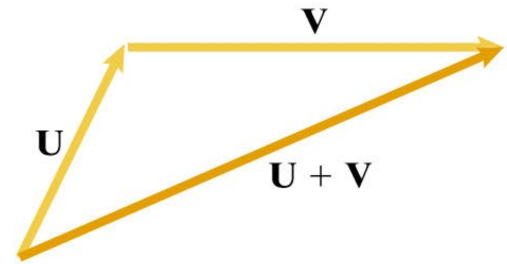
$$\vec{U} = (U_x, U_y, U_z)$$

$$\vec{V} = (V_x, V_y, V_z)$$

TRIANGULAR ADDITION:

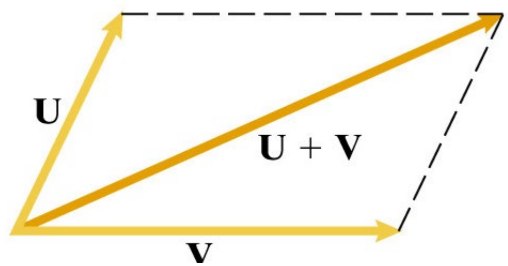


2. Attach the tail of $\vec{V}$ to the tip of $\vec{U}$



(U+V) is the vector from tail of $\vec{U}$ to the tip of $\vec{V}$

PARALLELOGRAM ADDITION:



2. Attach both tails and complete parallelogram with $\vec{U}$ and $\vec{V}$ as sides

ALGEBRAIC ADDITION:

$$\vec{U} + \vec{V} = (U_x + V_x, U_y + V_y, U_z + V_z)$$

3. To sum $\vec{U}$ and $\vec{V}$, add component by component

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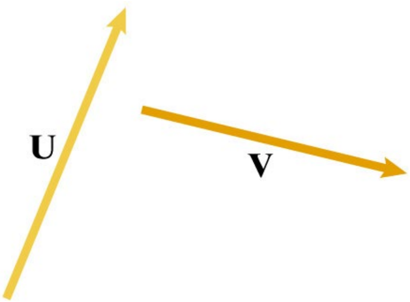
Reference: Trim, Ch.11, Section 11.3

Subtraction of Vectors



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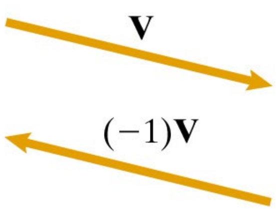
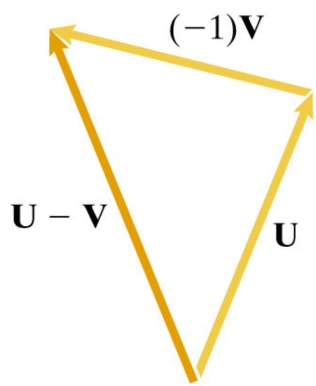


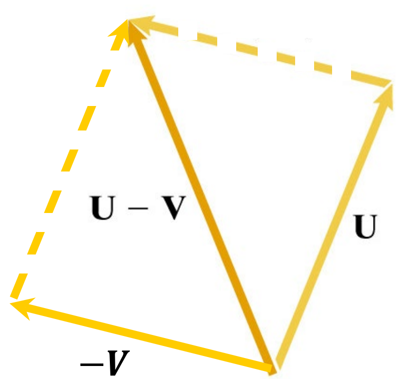
1. **Two Vectors $\vec{U}$ and $\vec{V}$**

$$\vec{U} = \left(U_x, U_y, U_z\right)$$
$$\vec{V} = \left(V_x, V_y, V_z\right)$$

TRIANGULAR SUBTRACTION:



PARALELLOGRAM SUBTRACTION:



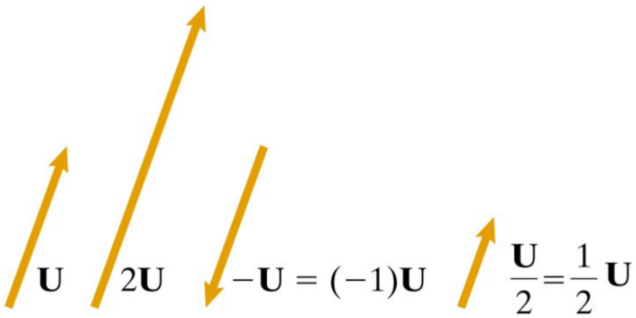
2. **Add Vectors $\vec{V}$ and $-\vec{V}$**

3. **$\vec{U} - \vec{V}$ is the sum of $\vec{U}$ and $-\vec{V}$**

$$\vec{U} - \vec{V} = \left(U_x - V_x, U_y - V_y, U_z - V_z\right)$$

Reference: Trim, Ch.11, Section 11.4

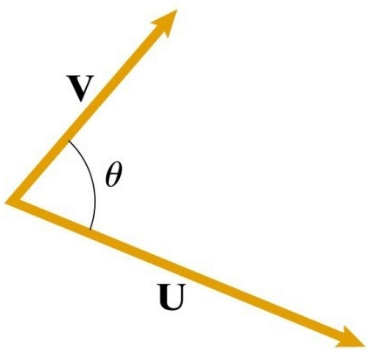
Scalar Products of Vectors



SCALAR MULTIPLICATION:

A vector $\vec{U}$ multiplied by a positive scalar value λ is obtained by multiplying each components of the vector by the scalar value:

$$\lambda\vec{U} = \lambda\left(U_x, U_y, U_z\right) = \left(\lambda U_x, \lambda U_y, \lambda U_z\right)$$



SCALAR PRODUCT (Dot Product):

The scalar product of two vectors $\vec{U}$ and $\vec{V}$ is defined as:

$$\vec{U} \cdot \vec{V} = |\vec{U}||\vec{V}|cos\ \theta \longrightarrow \textbf{SCALAR}$$

Where θ is the angle between $0 \leq \theta \leq \pi$ defined by the vector pair when the initial points coincide

<u>ANGLE BETWEEN TWO VECTORS</u>: $$cos\ \theta = \frac{\vec{U} \cdot \vec{V}}{|\vec{U}||\vec{V}|}$$

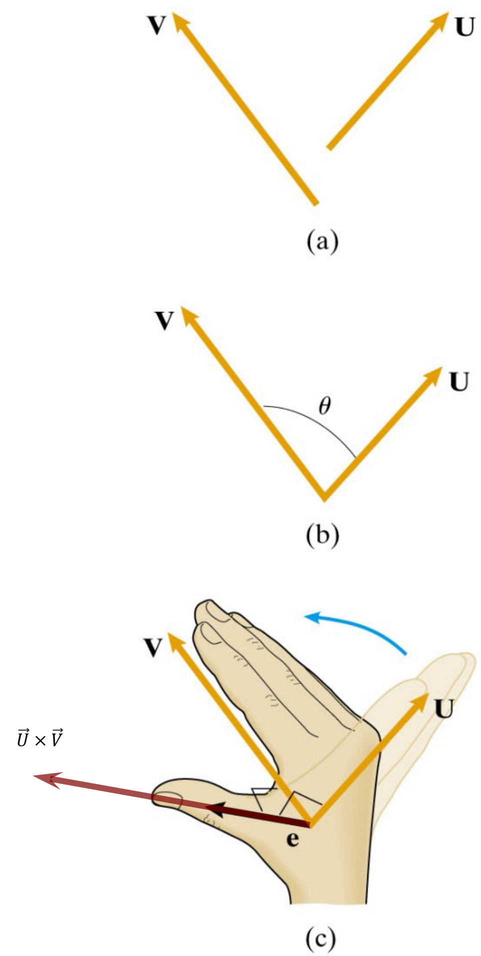
Reference: Trim, Ch.11, Section 11.4

Scalar Product: Useful Properties

Two Vectors $\vec{U}$ and $\vec{V}$	$\vec{U} = \left(U_x, U_y, U_z\right)$ $\quad \vec{V} = \left(V_x, V_y, V_z\right)$ $\quad \theta$: Angle in between
Scalar Product of $\vec{U}$ and $\vec{V}$	$\vec{U} \bullet \vec{V} = U_x V_x + U_y V_y + U_z V_z$
Commutative Law	$\vec{U} \bullet \vec{V} = \vec{V} \bullet \vec{U}$
Distributive Law	$\vec{U} \bullet (\vec{V} + \vec{W}) = \vec{U} \bullet \vec{V} + \vec{U} \bullet \vec{W}$ $(\vec{U} + \vec{V}) \bullet \vec{W} = \vec{U} \bullet \vec{W} + \vec{V} \bullet \vec{W}$ $(\vec{U} + \vec{V}) \bullet (\vec{W} + \vec{X}) = \vec{U} \bullet \vec{W} + \vec{U} \bullet \vec{X} + \vec{V} \bullet \vec{W} + \vec{V} \bullet \vec{X}$
Orthogonal Vectors	$\vec{U} \bullet \vec{V} = 0$ if an only if $\vec{U}$ and $\vec{V}$ are perpendicular $(\cos 90° = 0)$
Coincident Vectors	$\vec{U} = \vec{V}$ then $\theta = 0$ and $\cos\theta = 1 \rightarrow \vec{U} \cdot \vec{U} = \left\|\vec{U}\right\|^2$
Parallel Vectors	$\theta = 0$

Reference: Trim, Ch.11, Section 11.4

Products of Vectors



VECTOR PRODUCT (Cross Product):

a. Two Vectors U and V

$$\vec{U} = (U_x, U_y, U_z)$$

$$\vec{V} = (V_x, V_y, V_z)$$

b. The angle θ between the vector when placed tail to tail

Two vectors, $\vec{U}$ and $\vec{V}$, separated by an angle θ, form a plane

c. Unit vector $\hat{e}$ perpendicular to plane formed by U and V pointing in a direction dictated by the right-hand rule

The cross product is defined as:

$$\vec{U}\times\vec{V} = \hat{e}|\vec{U}||\vec{V}|sin\theta$$

Reference: Trim, Ch.11, Section 11.4

Vector Product: Useful Properties

Two Vectors $\vec{U}$ and $\vec{V}$

$$\vec{U} = (U_x, U_y, U_z) \qquad \vec{V} = (V_x, V_y, V_z)$$

Vector Product of $\vec{U}$ and $\vec{V}$

$$\vec{U} \times \vec{V} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ U_x & U_y & U_z \\ V_x & V_y & V_z \end{vmatrix}$$

$$\vec{U} \times \vec{V} = \hat{\imath}(U_yV_z - U_zV_y) - \hat{\jmath}(U_xV_z - U_zV_x) + \hat{k}(U_xV_y - U_yV_x)$$

The Vector Product is not commutative: $\vec{U} \times \vec{V} = -\vec{V} \times \vec{U}$

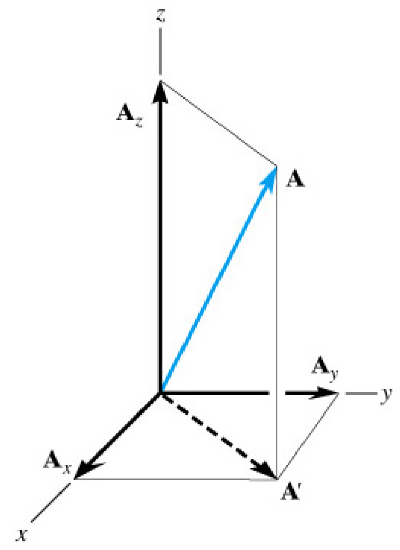
Distributive Law

$$\vec{U} \times (\vec{V} + \vec{W}) = \vec{U} \times \vec{V} + \vec{U} \times \vec{W}$$

$$(\vec{U} + \vec{V}) \times \vec{W} = \vec{U} \times \vec{W} + \vec{V} \times \vec{W}$$

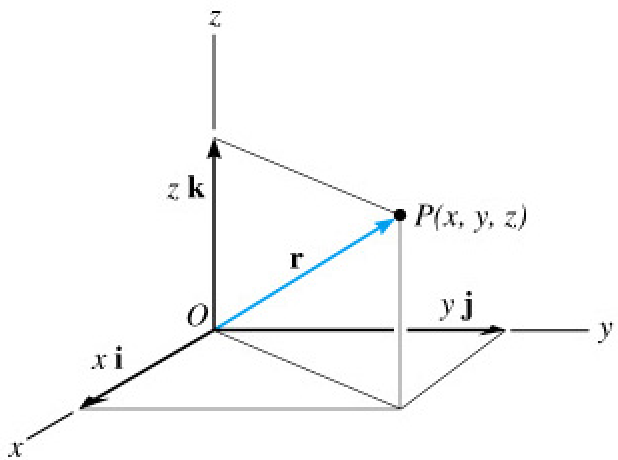
Rectangular Coordinates & Vectors

Rectangular Coordinates of a point in space are the directed distances from 3 fixed planes called the coordinate planes.

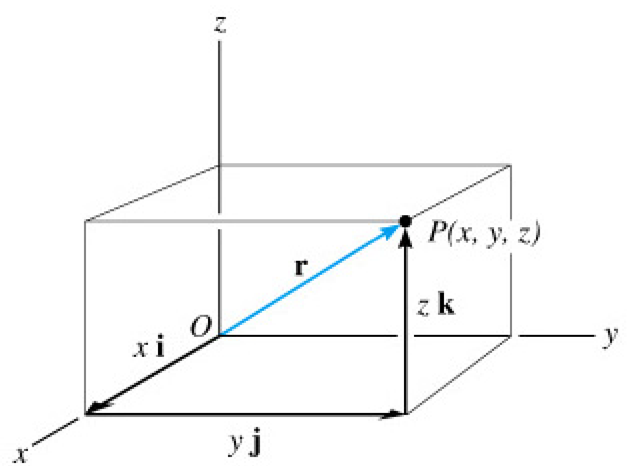


In a right-handed rectangular coordinates, point A is represented as $A = \left(A_x, A_y, A_z\right)$

Position of a Point in Rectangular Coordinates



Position vector in Cartesian Coordinates



Head to tail vector addition of components

POSITION VECTOR:

Fixed vector which locates a point in space relative to another point

In Rectangular Coordinates:

The position vector $\vec{r}$ extends from the origin 0, to point $P(x, y, z)$. Then,

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$$

Or :

$$\vec{r} = |\vec{r}|\hat{r}$$

$$\vec{r} = \left(\sqrt{x^2 + y^2 + z^2}\right)\hat{r}$$

Where $\hat{r}$ is the unit vector in the direction of $\vec{r}$. Therefore:

$$\hat{r} = \frac{\vec{r}}{|\vec{r}|}$$

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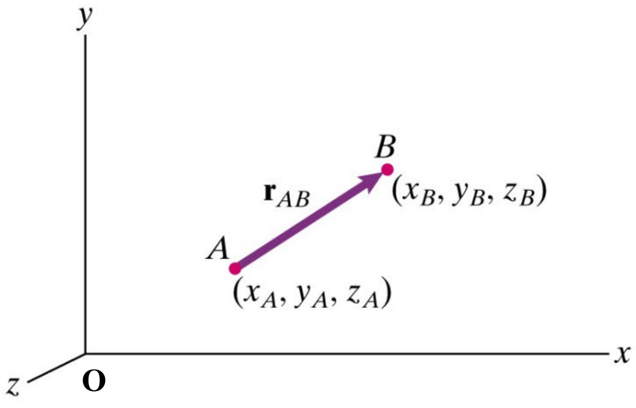
Vector Components in Rectangular Coordinates



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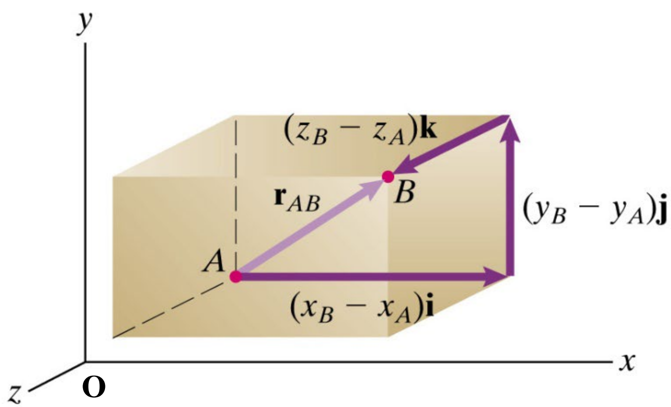

(b)

Given the vector from point A to B:

$$\vec{r}_{AB} = \vec{r}_{OB} - \vec{r}_{OA}$$

The components of the vector are given by:

x component: $x_B - x_A$

y component: $y_B - y_A$

z component: $z_B - z_A$

The vector is written as:

$$\vec{r}_{AB} = (x_B - x_A)\hat{\imath} + (y_B - y_A)\hat{\jmath} + (z_B - z_A)\hat{k}$$

<u>EQUALITY</u>:

Two vectors are equal if and only if they have the same components

Thank You